Adapting bioinformatics data systems in the era of foundational models: leveraging retrieval-augmented generation and low-resource large language models

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Short Report

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Abstract

We investigated how to leverage existing assets, such as curated life science database catalogs, in the era of information retrieval powered by large language models (LLMs). Although LLMs exhibit unprecedented information provision capabilities, they inherently suffer from hallucinations, which is an unavoidable limitation. Retrieval-augmented generation (RAG) is a promising approach to mitigate this issue. Furthermore, the analysis of personal data, such as human biological samples, must be conducted in an isolated environment, precluding the use of external Internet-based services. If a system that integrates LLMs and a RAG could be implemented within an isolated environment, it would significantly enhance research activities, including those involving personal data analysis. We evaluated the feasibility of using local LLMs and the effectiveness of the RAG score in reducing the incidence of hallucinations. Regarding the former, existing technologies such as those in Ollama suggest that local deployment is viable. For the latter, a rigorous selection of data sources for the RAG is essential. In particular, we found that establishing a well-structured repository of Japanese-language resources is crucial. Future challenges include optimizing the LLMs for this system and incorporating AI agent functionalities to enhance its overall performance.

Keywords: Foundational models, Large language models, Retrieval-augmented generation, In-context learning

1 Introduction

The National Institutes of Biomedical Innovation, Health, and Nutrition (NIBN), in collaboration with the Bioscience Database Center (NBDC), has systematically cataloged domestic bioinformatics data. This catalog has been integrated into a cross-search system utilizing the full-text search engine ElasticSearch. However, the landscape of information retrieval underwent a significant transformation with the introduction of ChatGPT in November 2022.

Although large-scale foundational models such as ChatGPT offer substantial convenience, their dependence on token occurrence probabilities within language models inherently leads to hallucinations[1], regardless of the accuracy of the pretraining data. Furthermore, the widespread adoption of these models poses the potential risk of undermining established efforts in bioinformatics data curation. To mitigate these challenges, we investigated the application of retrieval-augmented generation (RAG) to leverage curated bioinformatics data in conjunction with foundational language models.

Despite numerous studies demonstrating the effectiveness of RAG, a variety of methodologies have been proposed, and no consensus has been reached regarding the optimal approach for bioinformatics data retrieval. Consequently, extensive evaluation of these methodologies is required to determine their suitability in this domain.

A fully developed and user-friendly system is essential for widespread public adoption. One practical implementation strategy involves leveraging noncode platforms such as Dify[2], which streamlines deployment but is inherently constrained by user interfaces. Alternatively, a more flexible solution, albeit requiring Python programming expertise, involves integrating Streamlit with frameworks such as LangChain[3] or the LlamaIndex[4]. High-performance GPU environments were a prerequisite for the deployment of large language models (LLMs) until early 2024. However, advancements in model quantization techniques and the emergence of systems, such as Ollama[5], have facilitated the efficient operation of LLMs in resource-constrained environments. In addition, the recent introduction of Entropix technology is expected to further increase the computational efficiency of LLMs, thereby enabling their deployment with minimal resource requirements.

The utilization of LLMs locally deployed via Ollama, as opposed to commercially available public LLMs, has two significant implications. First, enhanced security is ensured. Our work frequently involves handling sensitive personal information such as human biological samples, which must not be exposed to the external environment. Even if commercial LLMs are bound by confidentiality agreements, most organizations prohibit the transmission of such data through external Internet connections. Second, operating in a closed environment without external Internet access restricts the ability to validate analytical methods or the plausibility of the obtained data in real time. However, local LLMs circumvent these limitations, enabling a more effective and secure data analysis and research progression.

Considering this, we aimed to evaluate the feasibility and performance of various LLMs currently available through Ollama in local environments. Specifically, we assessed the extent to which local LLMs can be utilized effectively. Furthermore, we investigated the implementation of RAG systems using these models and validated the data sources required to construct such RAG frameworks. This approach provided critical insights into the practical applicability of local LLMs in secure and restricted research settings.

2 Results

2.1 Survey of recent LLM environments

We used four LLMs from Ollama.

- Phi-4: Phi-4 is a 14B parameter, a state-of-the-art open model based on a combination of synthetic datasets, data from filtered public-domain websites, and acquired academic books, and Q and A datasets. The model parameters are 14B, and the file size is 9.1 GB [6].
- Gemma3: Gemma is a lightweight family of Google models built using Gemini technology. The Gemma3 models are multimodal, process text and images, and feature a 128 K context window with support for over 140 languages. The model parameters are 4B, and the file size is 3.3 GB [13].
- Qwen3: Qwen3 is the latest generation of LLMs in the Qwen series, offering a comprehensive suite of dense and mixture-of-experts models. The model parameters are 8B, and the file size is 5.2 GB [14].

The following zero-shot queries were performed for each LLM on the ollama, using the script shown in Fig 1. The query results are presented in Table 1.

Table 1 Zero-shot generation speed (words/s) for each model across different environments

Environment	Phi-4	Gemma3	Qwen3
Rocky Linux 8.1 (256GB) / RTX A6000 Ada (48GB)	60.13	113.21	42.11
Ubuntu 24.04 (64GB) / GeForce RTX 3070 (8GB)	9.06	73.49	53.01
Sequoia / Mac mini M4 (32GB)	8.06	23.45	14.51

```
import time
import requests
import json

# Ollama API endpoint
ollama_api_url = "http://localhost:11434/api/generate"  # Default endpoint

# Model name to use (must be loaded in Ollama)
model_name = "qwen3"  # Change this to the model name you want to measure on Ollama

# Prompt for text generation
prompt = "What do you think about Biomedical Linked Annotation Hackathon?"

# Maximum number of tokens to generate
max_tokens = 256

# Number of measurement runs
num_runs = 10

generation_times = []
output_lengths = []

for _ in range(num_runs):
    payload = {
        "prompt": prompt,
        "model": model_name,
        "stream": False,  # Disable streaming output
        "options": {
            "num_predict": max_tokens
        }
    }

    start_time = time.time()
    response = requests.post(ollama_api_url, json=payload)
    end_time = time.time()

    if response.status_code == 200:
        data = response.json()
        generation_time = end_time - start_time
        generation_times.append(generation_time)
        # Get approximate token count by counting words
        output_lengths.append(len(data['response'].split()))
    else:
        print(f"Error: {response.status_code} - {response.text}")
        exit()

# Calculate results
average_generation_time = sum(generation_times) / num_runs
total_generated_words = sum(output_lengths) # Note: This is word count
average_words_per_second = total_generated_words / sum(generation_times)

print(f"Ollama Model: {model_name}")
print(f"Prompt: '{prompt}'")
print(f"Max Tokens to Generate: {max_tokens}")
print(f"Number of Runs: {num_runs}")
print(f"Average Generation Time: {average_generation_time:.4f} seconds")
print(f"Total Generated Words (Approx. Tokens): {total_generated_words}")
print(f"Average Words/Second (Approx. Tokens/Second):
      f"{average_words_per_second:.2f} words/s")
```
Fig. 1 A Script for Quantitative Evaluation of Response Latency in each LLMs

2.2 Survey of RAG source

We chose the following resources listed in Table 2 as a candidates for RAGs

Table 2 Candidates of RAG source

Data source	URL
Integbio Database Catalog	https://catalog.integbio.jp/dbcatalog/en/
Manbyo-Dictionary	https://sociocom.naist.jp/manbyo-dic-en/
RDF Portal	https://rdfportal.org/
FANTOM5	https://fantom.gsc.riken.jp/5/
Gene Ontology	https://geneontology.org/
Disease Ontology	https://disease-ontology.org/
Drug Ontology	https://bioportal.bioontology.org/ontologies/DRON
ChEBI	https://www.ebi.ac.uk/chebi/
Human Phenotype Ontology	https://hpo.jax.org/
TargetMine	https://targetmine.mizuguchilab.org/

2.3 Validation of RAG's effectiveness

While the sources for Retrieval-Augmented Generation (RAG) were cited above, realizing its functionality necessitates the construction of a vector search mechanism. This study investigates the efficacy of in-context learning by embedding selected text into the prompt. The data utilized comprises Japanese content from https://scardahp.nibn.go.jp/, which details a system for proactively leading research and development during peacetime to rapidly advance vaccine development as a national strategy in the event of an infectious disease crisis. We queried the Gemma3 LLM about the 'Strategic Center of Biomedical Advanced Vaccine Research and Development for Preparedness and Response,' comparing responses with and without embedding this content in the prompt. It was presumed that this website was not yet incorporated into Qwen3's knowledge base, owing to its recent publication. Fig 2 was without ICL query, Fig 3 was with ICL query. The former, lacking requisite information, produced hallucinations; the latter, however, generated text in accordance with the facts.

3 Discussion

The use of lightweight inference engines such as Ollama has demonstrated that computational processes traditionally requiring extensive GPU resources can be executed using relatively modest hardware.

Based on experience, a system with a total memory capacity, comprising both main and GPU memories, equivalent to at least twice the size of the LLM, can ensure a seamless response. Although servers equipped with high-performance GPUs, such as A100 or A6000, may not be widely available, their use enables local processing without reliance on commercial generative AI services, such as ChatGPT, which is

```
**1. What is SCARDA?**

* **Established by the Biomedical Advanced Research and Development
  Authority (BARDA):
** SCARDA is a program *within* BARDA, which itself is a critical
  part of the U.S. Department of Health and Human Services.
* **A Public-Private Partnership:** It's fundamentally a collaborativ
  effort, bringing together government agencies, pharmaceutical companies,
  biotechnology firms, and research institutions.
* **Focus on Rapid Response:** SCARDA's core mission is to accelerate
  the development and manufacturing of vaccines against threats like
  pandemics, emerging infectious diseases, and bioterrorism. It'
  designed for *speed* { aiming to shorten the timeline from threat
  identification to vaccine availability.

(Truncated)
```

Fig. 2 Prompt to Gemma3 LLM about SCARDA without ICL

```
  **What is SCARDA?**

SCARDA (Strategic Center of Biomedical Advanced Vaccine Research
  and Development for Preparedness and Response) is a Japanese
  government-established organization designed to be the *central
  driver* of vaccine research and development, particularly in
  preparation for pandemic outbreaks.

**Key Features & Purpose:**

* **Established in 2022:** Created in March 2022, following
  a national strategy ("Vaccine Development and Production System
  Strengthening Strategy").
* **Strategic Funding:** Its primary role is to proactively identify
  and fund strategic vaccine research projects. It does this by
  gathering and analyzing extensive information on vaccine development.
* **Two Main Activities:**
    * **Funding:**  It directs research funding to projects
      deemed strategically important.
    * **Infrastructure Development:** It supports the creation of
      world-class research facilities dedicated to vaccine development.
    * **Coordination:** SCARDA acts as a central coordinating body,
      managing and overseeing  research efforts both during normal
      times and in the event of a pandemic.

(Truncated)
```

Fig. 3 Prompt to Gemma3 LLM about SCARDA with ICL
(Source: Content related to https://scardahp.nibn.go.jp/)

highly important. Furthermore, modern laptop computers increasingly feature dedicated GPUs, and even those without GPUs can be configured with 32 GB or more RAM, making them viable for LLM applications. Given the inherent challenge of hallucinations in LLMs, an essential issue remains: how to mitigate this phenomenon effectively to increase reliability and accuracy. Although we initially planned to assess the speed of Llama4, this evaluation was omitted. The decision was based on the substantial size of the LLM, which led us to anticipate that it would exceed the operational capacity of our available hardware: a GeForce 3070 (8 GB VRAM) and an Apple Metal system (32 GB unified memory).

This study demonstrates that instead of utilizing RAG, the direct embedding of arbitrary descriptions within prompts conditions the model's generative output for a given query on these embedded texts. In the absence of these embedded descriptions, queries generate entirely baseless and erroneous outputs (hallucinations). These findings indicate that existing well-curated databases can be effectively leveraged and remain valuable in the era of LLMs when utilized as a knowledge base within an RAG framework.

The utilization of the life science information described in the RDF and its associated ontologies as a source for an RAG appears to be a rational approach. By incorporating these resources into the RAG, the relationships among individual biological entities that are not inherently present in the LLM will be better captured, and the accuracy of terminology representation will be enhanced.

A data warehouse may seem unconventional as a source of RAG; however, TargetMine [8] integrates the aforementioned data as a drug discovery support system. The backend database of TargetMine is PostgreSQL, which enables a vector search using pgvector. In addition, InterMine, the framework underlying TargetMine, supports Ontop[9] , which was privately incorporated into TargetMine. Therefore, TargetMine is considered a viable source of RAG.

4 Future works

In the future, we must further investigate the following research directions.

4.1 Ongoing review of Japanese language resources for RAG and its evaluation

This investigation focused on leveraging the Japanese Life Science Database (LSDB) catalog within the context of LLMs. Although we identified several potential resources for use as RAG sources, the limited availability of Japanese-language data poses a challenge. To effectively mitigate hallucinations during Japanese text generation, a greater compilation of appropriate Japanese-language resources is needed. Various methods have been proposed to determine RAG scores. In this study, we focused on an implementation called the Medical GraphRAG[7]. Although we could verify the build process of the code available on GitHub, the repository did not contain the actual source code for the RAG. Because the primary objective of this research was to mitigate hallucinations in Japanese text generation via LLMs, evaluating the effectiveness of RAG in the context of Japanese language generation is essential.

4.2 Superiority evaluation of the ICL and RAG

ICL enhances the model performance by refining the structure and presentation of the input data without directly modifying the model parameters. Few-shot learning and chain-of-thought prompts are prime examples of such techniques. By contrast, RAG complements the model's knowledge by incorporating relevant information from external knowledge bases. Although distinct, both ICL and the RAG can be synergistically combined to address more complex tasks. This is particularly useful for LLMs with limited context windows, such as when domain-specific knowledge is lacking. By effectively combining ICL and RAG, we expect to achieve more advanced generative capabilities in LLM, such as Llama 3.X [12].

4.3 Deployment

This system is designed primarily for operation in onsite or closed environments rather than as an external Internet service. Consequently, developing easily deployed mechanisms is essential. Given that certain RAG sources may require specific acquisition procedures for installers, establishing a deployment framework that accommodates these requirements is crucial.

4.4 AI agent

The primary objective of a search system is to conduct preliminary investigations as a foundation for initiating research. Actively considering subsequent developments based on the search results can be highly beneficial. The AI Scientist developed by SakanaAI serves as an effective tool to support this objective. However, the Deep Research systems, released by Google and other AI-related companies, can be regarded as a more generalized system with similar capabilities. Some of these systems have been made available as open-source software on platforms such as GitHub [10] [11]. Their functionality as research tools can be further enhanced by integrating them into the backend.

4.5 Supervised fine-tuning via own literature

Supervised fine-tuning (SFT) is an approach aimed at mitigating hallucinations in LLMs by enhancing domain-specific expertise through additional training. Although SFT is effective, but its implementation often requires substantial GPU resources, posing a significant barrier to its widespread adoption. By contrast, the RAG algorithm incorporates external databases into the generative AI pipeline, offering a more resource-efficient alternative. Although research on the hallucination-mitigation effects of both SFTs and RAG systems is ongoing, SFTs are generally considered more effective. Therefore, this is a promising avenue for future research, provided that sufficient computational resources are available. When developing LLMs for public release, meticulous attention must be paid to licensing issues related to training data. However, for internal use within an organization, SFTs can be conducted using proprietary data without the same level of public scrutiny.

5 Conclusion

LLMs enable a more effective utilization of existing databases by leveraging accumulated knowledge and data. As LLMs and their associated technologies continue to advance rapidly, it is essential to remain informed about ongoing developments and integrate emerging innovations.

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